PathSense: Smart Cane Navigation

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Limitations of Traditional Canes

Key Challenges for Visually Impaired Users

Traditional white canes primarily detect ground-level obstacles, **missing** hazards at chest and head height while lacking depth awareness for potholes and curbs, limiting user safety during navigation.

Why It Matters

Importance of Addressing Navigation Challenges

India has one of the **largest visually impaired populations**, especially in Kerala, where uneven roads pose significant risks. Our device addresses critical gaps in obstacle detection and emergency response.

Our Solution: Enhanced Navigation Safety

01 **Side Detection**

Dual sensors identify obstacles on both sides.

02 **Depth Sensing**

Sensor at the tip detects height changes.

03 **Emergency Alert**

SOS button sends alerts during emergencies.

System Flow Overview

01

Sensing

Sensors continuously scan the path ahead and below.

02

Alerting

Vibration feedback guides the user's next action.

03

Processing

A microcontroller checks sensor values against thresholds.

04

Integration

Ultrasonic and depth sensors work in harmony.

What Makes PathSense Unique

01 **Coverage**

Detects forward, side, and ground depth obstacles.

02 **Emergency**

Integrated SOS alert for fall response included.

03 **Precision**

Dedicated sensors specifically target potholes and puddles.

Component Placement on the PathSense Cane

Handle

Microcontroller and SOS button housed neatly.

Grip

Vibration motor for immediate silent feedback.

Sensors

Ultrasonic and depth sensors ensure navigation safety.

Practical Strengths of PathSense

01 **Low Cost**

Estimated build cost is **affordable** for users.

02 **Proven Components**

All core parts are **well-documented** and accessible.

03 **Retrofit Design**

Attaches to standard canes, requiring **no custom builds**.

Next Steps

Sensor Logic

Finalizing and testing sensor logic in simulation ensures the functionality works as expected before moving to the physical prototype, minimizing potential issues during the build phase.

Component Sourcing

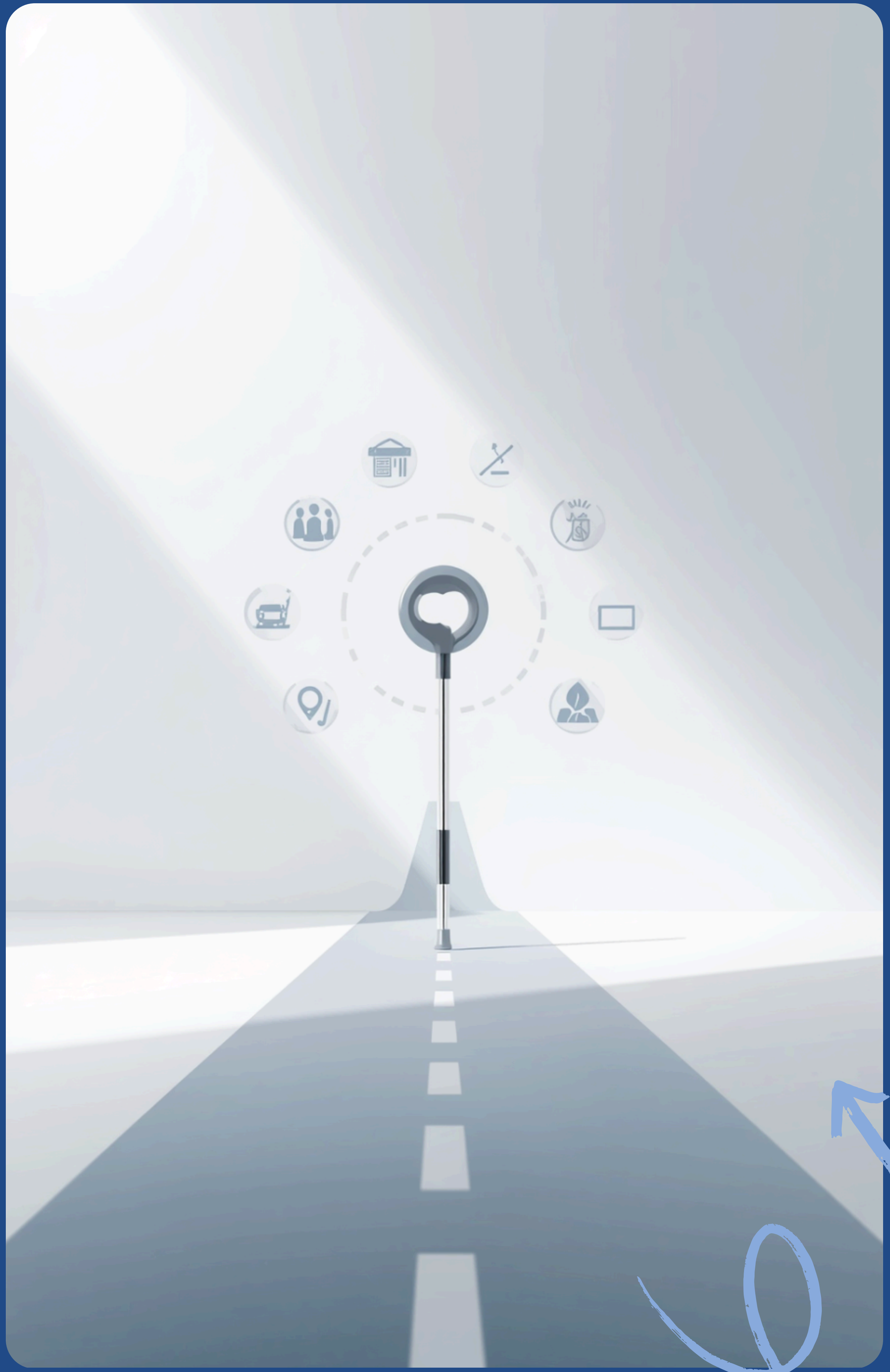
Sourcing components for the physical prototype will begin once the simulation is validated, allowing us to gather all necessary parts for assembly efficiently and effectively.

Field Testing

Conducting field tests on real pavement conditions will help us refine the sensor thresholds and ensure accurate obstacle detection and responses in varied environments.

Feature Exploration

Exploring SMS and GPS integration for the SOS feature will enhance user safety, providing reliable emergency communication options directly from the PathSense device during critical situations.



Thank You

PathSense: Confident Navigation for All